

THERMAL ENGINEERING - I

II B.TECH II Semester: MECHANICALENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME18	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OVERVIEW: This course will make students to understand principles of Internal combustion engines, study testing and performance of compressors.								
COURSE OUTCOMES: The course should enable the students to: 1. Demonstrate the basic Cycles and operation of IC engines. 2. Analyse the combustion process and also how it does affect the performance of the IC engines. 3. Describe procedures of testing and analyses the emission of IC engines. 4. Apply the modern developments and alternate fuels on IC engines. 5. Formulate and perform the maintenance and operation of Air compressors.								
UNIT-I	AIR STANDARD CYCLES AND IC ENGINES							
Air standard Cycles - Otto, Diesel and Dual cycle. IC Engine- Classification, Working principle, Valve and Port Timing Diagrams. Comparison of Air Standard and Actual Cycles, Time Loss Factor, Heat Loss Factor, Loss due to Rubbing Friction.								
UNIT-II	COMBUSTION IN IC ENGINES							
Combustion in S.I. Engines: Normal Combustion and abnormal combustion, Importance of flame speed and effect of engine variables, Type of Abnormal combustion, pre-ignition and knocking. Combustion in C.I. Engines: Four stages of combustion, Delay period and its importance, Effect of engine variables, Diesel Knock, Need for air movement.								
UNIT-III	IC ENGINE TESTING AND EMISSION							
Performance of IC Engines: Measurement and analysis of engine performance. Factors effecting efficiency. Variable speed, Air/Fuel ratio. Heat balance test. Engine Emissions: SI and CI engine emissions, Harmful effects, Emissions measurement methods, methods for controlling engine emissions.								
UNIT-IV	MODERN DEVELOPMENTS AND ALTERNATE FUELS							
Super Charging, Stratified charge engine, Dual-fuel engines, CRDI and HCCI concepts Alternate Fuels for IC Engines: Need for use of alternate fuels, Use of alcohol fuels, Biodiesel								
UNIT-V	AIR COMPRESSORS							
Compressors: Classification, positive displacement and roto dynamic machinery, Rotary blower, vane sealed compressor, principle of working, efficiency considerations.								

Reciprocating Air Compressors: Principle of operation, work required, Isothermal efficiency, volumetric efficiency, clearance, multi stage compression, under cooling, saving of work, minimum work condition for stage compression.

Text Books:

1. V.Ganesan (2011), I.C. Engines, 3rd edition, Tata Mc Graw-Hill, New Delhi, India.
2. R.K.Rajput (2011), Thermal Engineering, 18th edition, Lakshmi Publications, New Delhi, India.

Reference Books:

1. Mathur, Sharma (2008), IC Engines, 3rd edition, Dhanpat Rai & Sons, New Delhi, India.
2. Pulkrabek (2008), Engineering fundamentals of IC Engines, 2nd edition, Pearson Education, New Jersey.
3. Rudramoorthy (2003), Thermal Engineering, 5th edition, Tata McGraw-Hill, New Delhi, India.